DR: KABSHA …PROMETRIC&PEARSON VUECOURSE 00201064018621

$$\text{Molarity} = \frac{\text{no of mole}}{\text{Liter of solvent}} \qquad \text{no of mole} \rightarrow \frac{wt}{M.wt}$$

$$\text{Molality} = \frac{\text{no. of moles}}{\text{kg of solvent}}$$

$$\text{Osmolality} = \text{No of moles} \times \text{No of atom solvent} \xrightarrow{\times 1000} \text{milli osmol}$$

Dr/Kabsha

$$\text{meq} = \frac{wt\,(mg) \times \text{valency}}{M.wt}$$

$$t_{\frac{1}{2}} = \frac{0.693}{k_e} \quad \text{or} \quad 0.693 \times \frac{Vd}{Cl} \rightarrow Vd = \frac{Dose}{Cp_0}$$

$$C_{ss} = \frac{F \cdot M.D}{Cl \cdot \tau} \qquad T_{ss} = 4\text{–}5\ t_{\frac{1}{2}}$$

50%	→ 1
75%	→ 2
87.5%	→ 3
95%	→ 4.3
99%	→ 6.6

$t_{\frac{1}{2}}$

$$\text{child dose} = \frac{BSA_{child}}{1.73} \times \text{dose adult}$$

$$\text{Plasma osmolality} = 2\,Na + \frac{glucose}{18} + \frac{BUN}{2.8}$$

$$AUC = \frac{Dose}{Cl} \qquad (Cl = k\,Vd)$$

$$\frac{AUC_{oral} \times Dose_{IV}}{AUC_{IV} \times Dose_{oral}} = F$$

$$dp/min = \frac{ml \times dp/ml}{min}$$

Rate of Flow

$$\text{phenytoin correction (ppm, mg/L)} = \frac{\text{observed (mg/L)}}{(0.2 \times \text{albumin (g/dl)}) + 0.1}$$

$$Ca\ (mg/dl) = \text{Total Ca (mg/dl)} + 0.8\,(4 - \text{Alb (g/dl)})$$

كثافة $D = \frac{wt}{Vol}$ وزن / حجم

$$CrCl = \frac{(140 - \text{age (year)}) \times wt\,(kg)}{Scr \times 72}$$

Female × 0.85

$$LD = \frac{Vd \times Cp_{ss}}{S \times F}$$

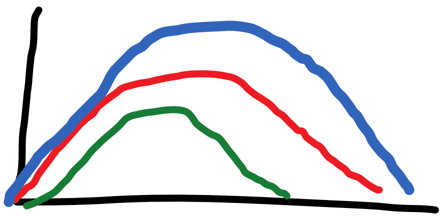
S → Salt factor

F → bioavailability

Bioavilabilty: rate & extent of abs ..& amount of drug reach to systemic circulation unchanged

Highest bioavilabilty : iv rout

High bioavilabilty =

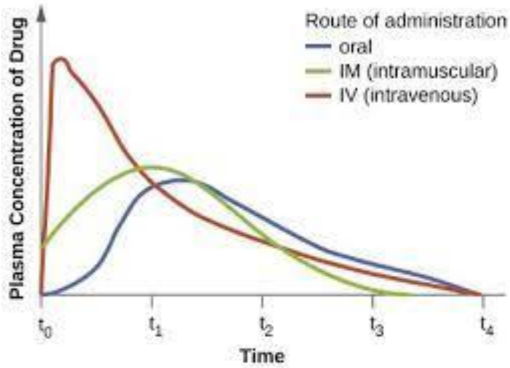


AUC= 30

AUC= 30

AUC= 25

.



84- Results data from a bioavailability study of four different dosage forms of an antibiotic drug is shown (see table). The volunteers fasted for 12 hours prior to taking the drug products. Urine samples were collected up to 72 hours after the administration of the drug to obtain the maximum urinary drug excretion.

Drug Dosage From	Dose (mg/kg)	Cumulative Urinary Drug Excretion 0-72 h
IV solution	0.2	20
Oral solution	4	380
Oral tablet	4	340
Oral capsule	4	360

Which of the following is the absolute bioavailability of the tablet dosage form?

- A. 80%
- B. 85%
- C. 90%
- D. 95%

2. If we give 250 ml of a drug and the area under curve was 112mg/hr/L and
after that we give 500 ml and the area under curve was 56 mg/hr/ L
The bioavilability decreased by

A-25%

b-50%

c-75%

Answer:

250ml 112

500 ml X

X= 122×500 / 250 = 224

But real auc was = 56

So the bioavilability dcrease from 224 to 56

That mean decrease by 224-56 =168

As a percent 168/224 X 100 =75%

3. **drug A taken IV and drug B taken orally**

the AUC of A =300 and Auc of b =225

what is biovalbility of drug

A. 85%

B. 90%

C. 75%

D. 80%

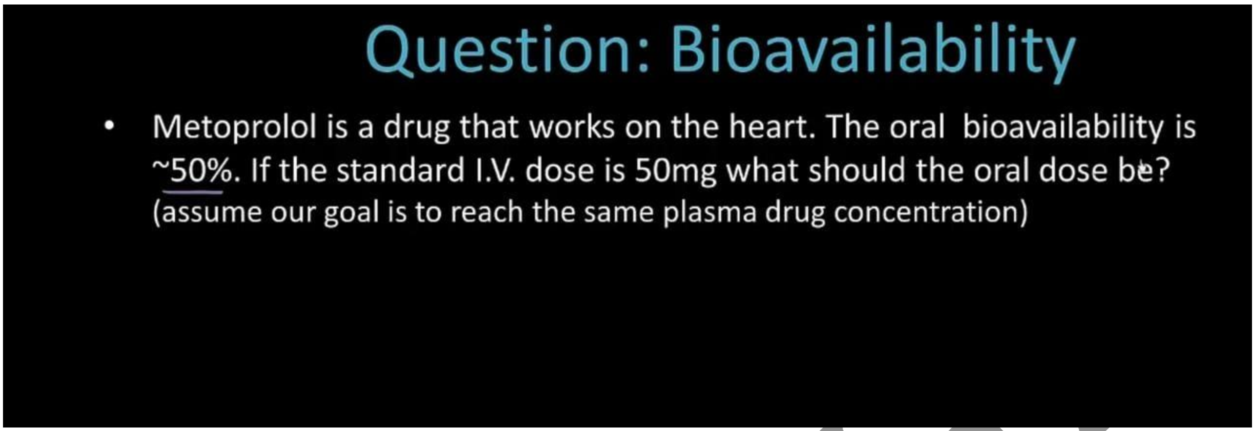
Answer:

Bioavailability= auc oral /auc iv ×100 = 225/300 × 100 = 75%

use the equation to calculate the bioavailability of a penicillin given as IV versus given as a tablet. Let's say that the AUC of 25 mg penicillin after administered as IV is 100 mg.h/L and that the AUC is 60 mg.h/L after giving a tablet containing 100 mg penicillin. What then is the absolute bioavailability of the penicillin?

$$F_{abs} = 100 \cdot \frac{60mg\ h/l \cdot 25\ mg}{100\ mg\ h/l \cdot 100mg} = 15\ \%$$

.

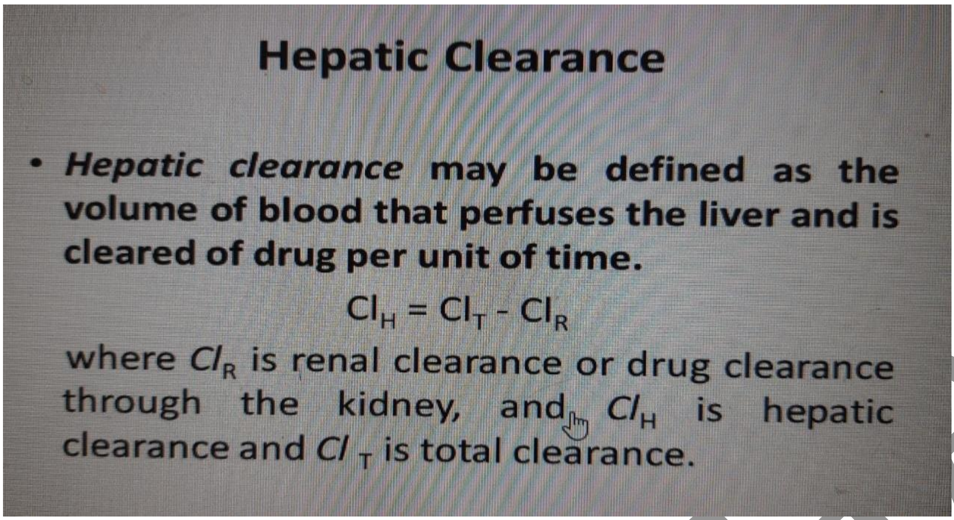


Ciproflixacin iv 400mg

تعادل كام **ciprofloxacin oral**

Bioavailabilty 80%

- **320 mg**
- **400 mg**
- **450 mg**
- **500 mg**

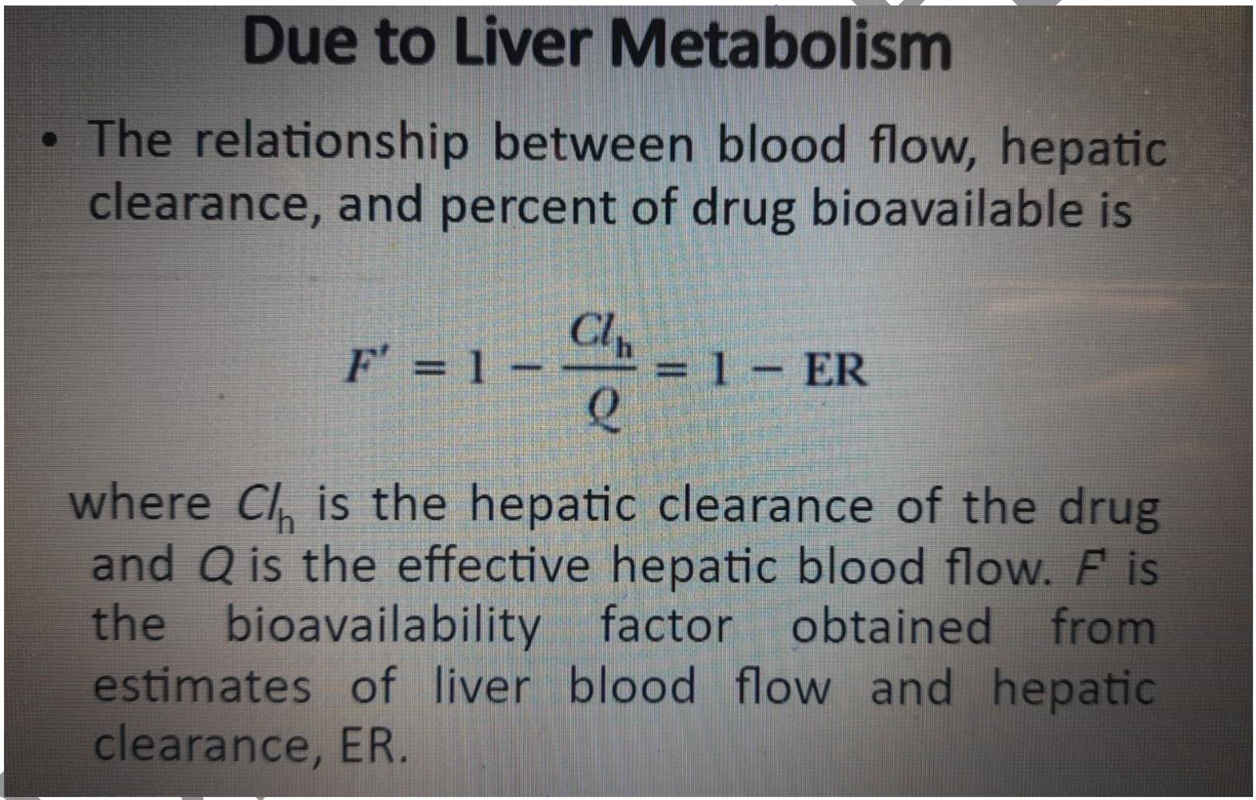


Hepatic Clearance

- ***Hepatic clearance*** **may be defined as the volume of blood that perfuses the liver and is cleared of drug per unit of time.**

$$Cl_H = Cl_T - Cl_R$$

where Cl_R is renal clearance or drug clearance through the kidney, and Cl_H is hepatic clearance and Cl_T is total clearance.



Due to Liver Metabolism

- The relationship between blood flow, hepatic clearance, and percent of drug bioavailable is

$$F' = 1 - \frac{Cl_h}{Q} = 1 - \mathrm{ER}$$

where Cl_h is the hepatic clearance of the drug and Q is the effective hepatic blood flow. F' is the bioavailability factor obtained from estimates of liver blood flow and hepatic clearance, ER.

Calculate bioavailability if hepatic blood flow is 1500 ml/min and hepatic clearance is 200 ml/min?

a) **0.6**
b) **0.90**
c) **0.87**
d) **0.78**

- The **volume of distribution** (V_D)

$$\text{Volume of distribution: } = \frac{\text{[amount of drug in the body]}}{\text{[serum concentration]}}$$

Interpretation of Vd in a 70 kg patient :

If Vd = 3 - 4 L, drug is mainly localized in plasma e.g. bound to plasma albumin or large MW.

If Vd = 10 - 12 L , then drug is localized in ECF i.e. it is water soluble and poorly enter cells

If Vd = 20 L , then drug is lipid soluble , and partially enter into cells across cell membranes of tissues

If Vd = 40 L , then drug is enough lipid soluble to be uniformly distributed in total body fluid e.g. alcohol

If Vd = > 42 L e.g.100 L, drug is **highly lipid soluble & is stored in some tissues** e.g. Digoxin Vd = 300 L. **Because of this large Vd for stored drugs, Vd is named : apparent i.e. aVd.**

Hemodialysis is not clinically effective to remove these stored drugs from body in cases of poisoning.

A drug has a volume of distribution of 35L in a 70 Kg man.Which of the following best describes Its distribution?

A. it is bound to DNA

B. it is dissolved in lipids

C. it has low bloavailability

D. it is mostly distributed in plasma

Loading dose & Maintainance dose

Loading dose

$$LD = \frac{C_p \times V_d}{F}$$

Cp = target plasma concentration
Vd = volume of distribution for the drug in that person
F = bioavailability for drug for route of administration being used

Maintenance dose:

$$MD = \frac{C_p \times CL \times \tau}{F}$$

$$CL = V_d \times k_e$$

$$k_e = \frac{0.693}{t_{\frac{1}{2}}}$$ k_e = elimination rate constant; $t_{½}$ = half-life

- drug has volume of distribution 15 liter and oral bioavailability 0.6 What is theloading dose in mg required to reach 0.8mg/ml steady state concentration?

A. **20**
B. **12**
C. **2**
D. **0.2**

2. A 70 kg man with atrial fibrillation. He requires an oral loading dose of digoxin to achieve a plasma digoxin concentration of 1.5 mcg/L.
Volume of distribution = 6 L/kg
Salt factor = 1
Bioavailability = 0.7
What is the oral loading dose of digoxin this patient requires?

6- Man weigh 78kg, he is taking 1g Vancomycin IV every 12 hr. Volume of distribution =1L/kg... t half = 8 hrs. Calculate Css (concentration at steady state).

Css= [F*MD] / [CL*T]

CL= [0.693*VD] /t half …. VD= 1L*78kg = 78L.

CL= [0.693*78] /8 CL=6.75 L/hr.

Css= [1*1g] / [6.75*12] ….. = 1/81

Css= 0.012g = 12mg.

1.10 A 55-year-old male patient (70 kg) is going to be treated with an experimental drug, Drug X, for an irregular heart rhythm. If the V_d is 1 L/kg and the desired steady-state plasma concentration is 2.5 mg/L, which of the following is the most appropriate intravenous loading dose for Drug X?

A. 175 mg.
B. 70 mg.
C. 28 mg.
D. 10 mg.
E. 1 mg.

15 A 40-year-old male patient (70 kg) was recently diagnosed with infection involving methicillin-resistant *S. aureus*. He received 2000 mg of vancomycin as an IV loading dose. The peak plasma concentration of vancomycin was reported to be 28.5 mg/L. The apparent volume of distribution is:

A. 1 L/kg.
B. 10 L/kg.
C. 7 L/kg.
D. 70 L/kg.
E. 14 L/kg.

wt = 75 kg

aminophyllin dose = 250 mg by IV

Vd = 0.5 L/kg

calculate serum conc

u must remember $F = 1$, $S = 0.8$

Then use $Vd = \frac{Dose}{Cp}$

$\therefore Cp = \frac{Dose \times \boxed{F \times S}}{Vd}$

remember $Vd = 0.5 \times 75 =$

- Patient came to clinic e asthma weight 78kg
Taking theophylline
T1/2 8.7
VD 0.5L/kg
Calculate infusion rate?
If steady state is 10mg/L
a) 31

- A 1000-mg dose of a drug was given intravenously to a 25-year-old male patient weighing 60 Kg. the initial plasma concentration of the drug was found to be 10 mg/L, and the elimination rate constant was 0.1 h. Assuming linear kinetics, what would be the total body clearance?

A.10 litres/hour

B.100 liters/hour

C. 150 liters/ hour

D. 250 litres/hour

I. What is the AUC of IV Ceftriaxone considering half-life is 6 hours and initial concentration is 30 mg. $259.7\ mg.h/L$

To calculate area-under the curve is:

$$AUC = \frac{C(0)}{Ke}$$

$$Ke = \frac{0.693}{half-life} = \frac{0.693}{6\ hours} = 0.1155\ h^{-1}$$

$$AUC = \frac{C(0)}{Ke} = \frac{30\ mg/L}{0.1155} = 259.7\ mg.h/L$$

حساب الكميات

Per dose?? Per day?? Per total duration?

6- The pharmacy receives a prescription for labetalol 800 mg twice daily for three days. The available strength of labetalol in the pharmacy is 200 mg.

Which of the following is the number of 200 mg labetalol tablet needed to prepare the above prescriptions for the total duration?

- A. 8
- B. 12
- C. 16
- D. 24

800 x2 = 1600 mg

1600 mg ÷ 200 mg = $\boxed{8}$

The Answer is (D)

The total dose for 3 days = {(800 mg x 2) x 3}
= 4800 mg

200 mg in 1 tablet
4800 mg in x tablets
X tablets ={ (4800 x 1) / 200 }
= 24 tablets

24- The weekly dose of docetaxel is 40 mg/m².

How many milliliters of docetaxel solution, 160 mg/16 ml would a patient measuring 1.5 m² need every week?

- A. 4 ml
- B. 6 ml
- C. 8 ml
- D. 10 ml

The Answer is (B)

40 mg for 1 m²
X mg for 1.5 m²
X mg = { (1.5 x 40) / 1 }
= 60 mg
160 mg in 16 ml
60 mg in x ml
X ml = { (60 x 16) / 160 }
= 6 ml

BSA Calculation

Mosteller Formula

$$\text{BSA } (m^2) = \sqrt{\frac{[\text{height (cm) x weight (kg)}]}{3600}}$$

www.nursingcenter.com

IBW male = 50+2.3 (inch -60) or ht in cm -100 for men

IBW female = 45+2.3(inch -60) or ht in cm – 105 fro female

(48)
An adult female patient with weight 80 kg and 160 cm hight.
How do you categorize this patient weight based on the following classification:

BMI	Weight Category
<18.5	Underweight
18.5-24.9	Normal
25-29.9	Overweight
>30	Obese

A. obese
B. normal
C. overweight
D. underweight

Weigh History

Usual Weight

$$\text{\% of UBW} = \frac{\text{Current Weight}}{\text{Usual Weight}} \times 100$$

Mild malnutrition: 85% -95%
Moderate malnutrition: 75%- 84%
Severe malnutrition: <74 %

36- A 6-year-old child weighing 28 kg came to the clinic with viral infection . The physician prescribed acyclovir at a dose of 20 mg/kg four times a day. Acyclovir is available as 200 mg/5 ml oral suspension.

How many milliliters would be administered per day ?

- A. 10
- B. 14
- C. 44
- D. 56

The Answer is (D)

The dose for 1 day for 1 kg

20 mg x 4 times = 80 mg
80 mg for 1 kg
X mg for 28 kg
X mg = { (28 x 80) / 1 }
= 2240 mg for one day

The stock

200 mg in 5 ml
2240 mg in X ml
X ml = { (2240 x 5) / 200 } = **56 ml**

54- A 50year-old girl weighing 20 kg is brought to the clinic with Lyme disease.

The physician prescribed cefuroxime 12.5 mg/kg twice daily for 14 days. The pharmacist has cefuroxime powder for suspension which after reconstitution with water forms 100 ml suspension with concentration of 125 mg /5 ml.

How many bottles would the pharmacist dispense ?

- A. 1
- B. 2
- C. 3
- D. 4

The Answer is (C)

12.5 x 2 x 14 = 350 mg for 1 kg
350 mg x 20 = 7000 mg
The bottle is 125 mg in 5 ml
7000 mg in X ml
X ml = 280 ml
One bottle = 100 ml
X bottle = 280 ml
X bottle = 3 bottle

56- A child weighing 25 kg is prescribed acetazolamide 2.5 mg/kg every 12 hours. Acetazolamide is available in a liquid dosage form with strength of 25 mg /ml.

How many milliliter would the patient be administered per dose?

- A. 2.5
- B. 5.0
- C. 7.5
- D. 10

The Answer is (A)

One dose = 2.5 x 25 = 62.5 mg

25 mg in 1 ml
62.5 mg in X ml
X ml = 2.5 ml

14- A prescription order asks for compounding dimenhydrinate syrup of strength of 12.5 mg/5 mL.

How many 50 mg tablets of dimenhydrinate is needed to prepare a 60 mL of the syrup?

- A. 1
- B. 2
- C. 3
- D. 6

12.5 mg —— 5 ml

6

The Answer is (C)

12.5 mg in 5 ml
X mg in 60 ml
X mg = { (60 x 12.5) / 5 }
= 150 mg

50 mg tablets in 1 stock
150 mg in x stock
X = { (150 x 1) / 50 }
= 3

. A Patient weighting 80 Kg is supposed to receive a drug at a dose of 2mg/kg/day. What is the dose that the patient should take for each day:

A. 80 mg .....

B. 160 mg .....

C. 240 mg .....

D. 320 mg .....

E. 400 mg

Kg=2.2 pound (ib)

1 inch= 2.54 cm

1 foot = 12 inches

1 gallon = 3.79 liters

Tea spoonful ((tsp)) = 5 ml

Table spoonfull ((tbsp)) = 15 ml

- **PPM = Part Per Milion = mg / L**
- **10% w/w = 10 gm in 90 gm ((total wt. = 100 gm)) --------- w/w = gm in gm**

- **10% w/v = 10 gm in 100 ml ((total voume = 100 ml)) --------- w/v = gm in ml**
- **10% v/v = 10 ml in 90 ml ((total voume = 100 ml)) ----------- v/v = ml in ml**

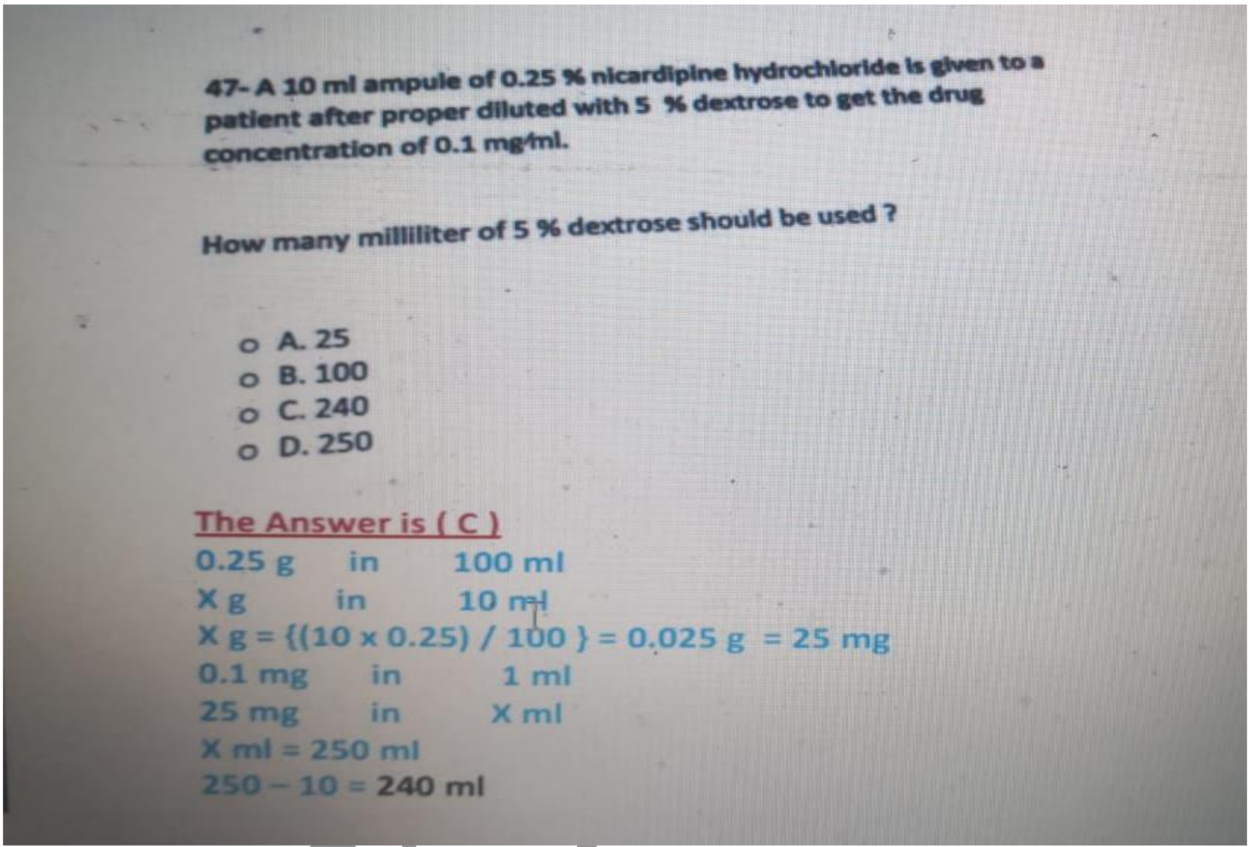
Measure of Length:

The meter (m) is the fundamental unit of length in the metric system.

1 kilometer	Km OR km	1000 meter = 10^3 meter
1 hectometer	Hm OR hm	100 meter = 10^2 meter
1 decameter	Dm	10 meter
1 decimeter	dm	0.1 meter = 10^{-1} meter
1 centimeter	cm	0.01 meter = 10^{-2} meter
1 millimeter	mm	0.001 meter = 10^{-3} meter
1 micrometer	µm	0.000 001 meter = 10^{-6} meter
1 nanometer	nm	0.000 000 001 meter = 10^{-9} meter

85 micro meter = ?? cm

Express 5 PPM of iron in water as a percentage.?



47- A 10 ml ampule of 0.25 % nicardipine hydrochloride is given to a patient after proper diluted with 5 % dextrose to get the drug concentration of 0.1 mg/ml.

How many milliliter of 5 % dextrose should be used ?

- A. 25
- B. 100
- C. 240
- D. 250

The Answer is (C)

0.25 g in 100 ml

X g in 10 ml

X g = {(10 x 0.25) / 100 } = 0.025 g = 25 mg

0.1 mg in 1 ml

25 mg in X ml

X ml = 250 ml

250 – 10 = 240 ml

1- amount of drug is 5 mg in 1 ml what the amount of drug in 1 tsp in microgram

a) 5mcg

b) 25 mcg

c) 500 mcg

d) 2500 mcg

e) 25000 mcg

5 mg → 1ml

?? mg → 5ml (tsp)

Answer

1 tsp = 5 ml

5 mg …. 1 ml

X mg …. 5 ml

X = 5 x 5/1 = 25 mg = (25 x 1000) 25000 mcg

2-An elixir contains 0.1 mg of drug X per ml. HOW many micrograms are

here in one tsp of the elixir

A. 0.0005 micrograms

B. 0.5 micrograms

C. 500 micrograms

D. 5 micrograms

E. 1500 micrograms

$mg \xrightarrow{\times 1000} micro$

$gm \xrightarrow{\times 1000} mg \xrightarrow{\times 1000} micro$

3-5ml of injection that conc. 0.4% w / v calculate the amount of drug?

a-0.2mg

b-2mg

c-200mg

d-2000mg

e-20mg

0.4 % mean ?

Answer:

0.4 gm ... 100 ml

X gm ... 5 ml

X = 5 x 0.4/100 = 0.02 gm = (0.02 x 1000) = 20 mg

14- A prescription order asks for compounding dimenhydrinate syrup of strength of 12.5 mg/5 mL.

How many 50 mg tablets of dimenhydrinate is needed to prepare a 60 mL of the syrup?

12.5 mg —— 5 ml

6

- A. 1
- B. 2
- C. 3
- D. 6

The Answer is (C)

12.5 mg in 5 ml

X mg in 60 ml

X mg = { (60 x 12.5) / 5 }

= 150 mg

50 mg tablets in 1 stock

150 mg in x stock

X = { (150 x 1) / 50 }

= 3

2- A pharmacy intern is preparing a solution containing 8.4 g of drug from an available stock solution labeled as 20% w/v.

Which of the following is the volume he must take from 20% w/v solution?

- A. 4.2 mL
- B. 8.4 mL
- C. 42 mL
- D. 84 mL

Answer (C)

20 g in 100 ml

8.4 g in X ml

X ml = (8.4 x 100) / 20

= 42 ml

6. HOW can prepare 100 ml of 12% MgCl by taking?

a-12ml of MGCL dissolve in 100 ml water

b-12 gm of MGCL dissolve in 100 ml water

c-12ml of MGCL dissolve in 1000 ml water

d-90.5 ml of MGCL dissolve in 100 ml water

7. A bag containing 250 ml of 25000 IU heparin The patient weigh 70 kg should recieve 10 IU/kg/hr...calculate the amount in ml the the patient should recieve in one hour...

Answer:

10 iu for 1 kg
X iu for 70 kg
X = 70 x 10 /1 = 700 iu
250 ml of 25000 iu
X ml of 700 iu
X = 700 x 250 / 25000 = 7 ml

8- How need prepare benzacainamid conc. 1:1000 ,30cc of benzocainamid solution?

a-30 mg

b-50 mg

c-80 mg

d-100 mg

e-130 mg

Note : cc = cubic centimeter = cm3= ml

Answer :

1 gm ----- 1000 ml

X gm ----- 30 ml

X = 30 x 1 / 1000 = 0.03 gm = 30 mg

9. Patient takes dose 20 mg/kg/day what is the dose if patient weight 60 pound ?

545 mg/day

Answer:

you have to know .. 1 kg = 2.2 pound (lb)

20 mg -------- 2.2 lb

X mg ---------- 60

X = 60 x 20 / 2.2 = 545.45 mg/day

Molarity : & osmolalty

1- A solution is made by dissolving 17.52 g of NaCl (m.wt = 57) exactly 2000 ml.

What is the molarity of this solution?

$\frac{\text{no of mole}}{L}$ → $\frac{\text{wt } 17.52 \text{ gm}}{\text{M.wt} \rightarrow 57}$

2000 mL → 2L ÷1000

a- 3.33

b- 0.15

c- 1.60

d-3.00 x 10 -4

e-1.6x10 -4

Answer :

Molarity=mole/volume (L)

1 Mole=molecular weight of subs. In 1 grams

No of Moles = wt / Mwt

So, Mole=17.52/57=0.307

So, Morality=0.307/2=0.153

2- if we have 0.8687g cacl2 (m,wt =111) in 500 ml solvent , denisty of the solvent is 0.95 g\cm3 ....Find the molality

a- 0.0165 Molal

b- 0.0156 Molal

c- 0.0165 m

d- 0.0156 m

no of moles → $\frac{wt}{M.wt} = \frac{0.8687\ g}{111}$ = 0.007 mole

kg → x

$D = \frac{mass}{vol}$ → 475 g

0.95 500

÷1000

0.475 kg

Answer :

Moles = mass/m.wt = 0.8687 / 111 = 0.007

Weight = density × volume = 0.95 × 500 = 475 gm = 0.475 kg

Molality = moles / kg of solvent = 0.007/0.475 = 0.0165 molal

---

0.9 % w/v

3-How many mOsm are present in 1 liter of sodium chloride injection (normal saline)

(Mwt: sodium chloride= 58.5) ?

0.9 gm → 100 ml

?? gm → 1000 ml = 1 L

→ 9 gm

308 mosm

Answer :

 Note ; normally conc. of NaCl injection = 0.9%

that means 0.9 gm in 100 ml ….. that means 9 gm in 1 L

osmol = no of mole x no of atom (solvent)

 Step 1.

$\frac{9\ wt}{58.5\ Mwt} \times 2$

= 0.308 osmol

x1000

308 mosm

millimoles = wt (gm) / Mwt (gm) × 1000 = 9 /58.5 ×1000 = 154

Note ; millimole = wt (mg) / Mwt (gm)

 Step 2.

mOsm = millimoles x no. of dissosation particles =154 × 2 =308 mosm

4-A solution contains 448 mg of KCl (MW=74.5) and 468 mg of NaCl (MW =58.5) in 500mL. What is the osmolar conc. of this solution ?

0.056 Osm/l

Answer :

▯ For (KCl)

0.448 gm in 500ml

X gm in 1000 ml …… X= 0.896 gm

moles= 0.896/74.5 = 0.012

Osm= moles × no. of dissosation particles =0.012 × 2= 0.024

▯ For NaCl

0.468 gm in 500 ml

X gm in 1000 ml ….. X= 0.936 gm

moles= 0.936 /58.5 = 0.016

Osm= 0.016 × 2= 0.032

▯ Total osmalar conc. of sol. = 0.032 + 0.024 = 0.056 Osm/l

5.For 1 litre of NaCl 3% calculate the osmolarity m.wt=58.5

1026

Answer:

3% means 3gm in 100 ml … that means 30gm in 1L

No. of moles = wt / Mwt = 30 / 58.5 = 0.513 mole

Osm = no. of mole × no. of dissosation particles = 0.513 × 2 = 1.026

1.026 x 1000 = 1026 mosm

6. The Molal concentration of 0.559 M solution is ;(Mwt=331.23 g/mol) (density of solution =1.157g/ml)

a-1.882

b-0.882

c-0.559

d-0.575

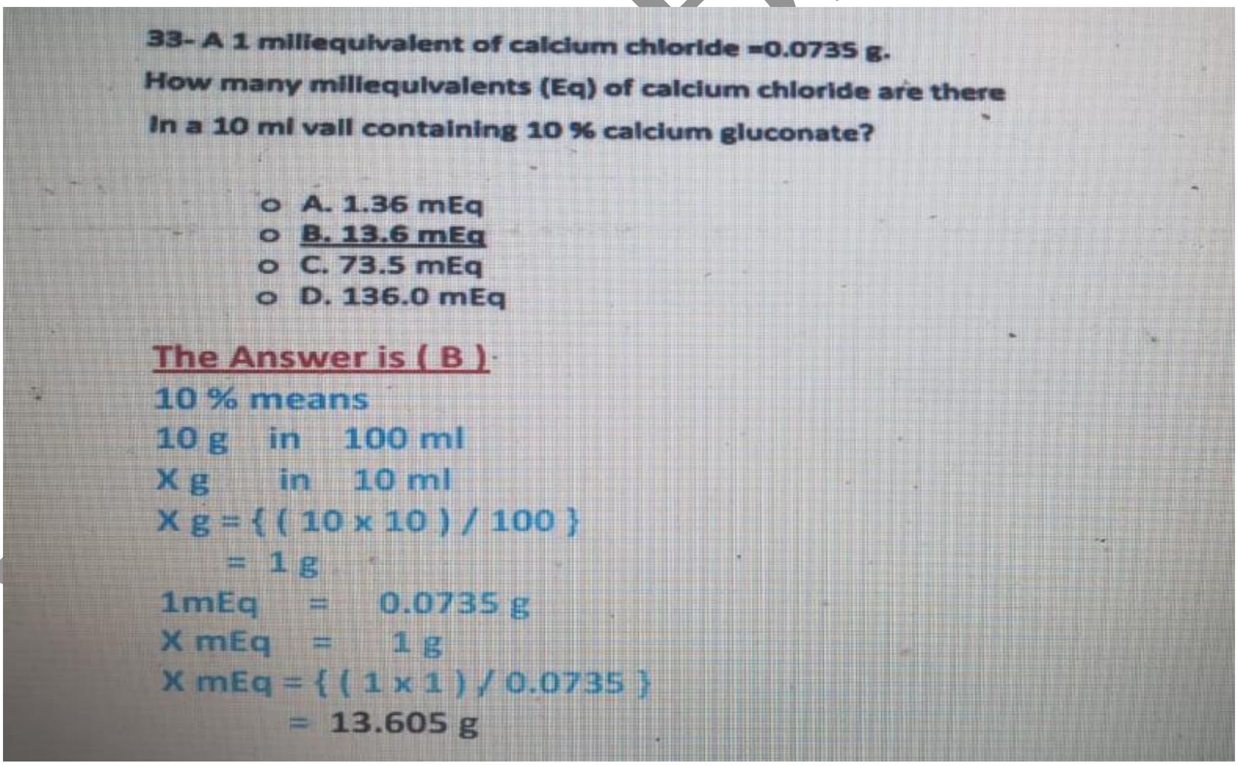
Answer :

Mass = moles × Mwt = 0.559 × 331.23 = 185.15 gm

wt of solution = Volume × Destiny = 1000 ml × 1.157=1157 gm

so wt of solvent = 1157 - 185.15 = 971.85gm = 0.971 kg

molality = moles / kg of solvent = 0.559 / 0.971= 0.575 molal



33- A 1 miliequivalent of calcium chloride =0.0735 g.
How many milliequivalents (Eq) of calcium chloride are there In a 10 ml vail containing 10 % calcium gluconate?

- A. 1.36 mEq
- B. 13.6 mEq
- C. 73.5 mEq
- D. 136.0 mEq

The Answer is (B)

10 % means

10 g in 100 ml

X g in 10 ml

X g = { (10 x 10) / 100 }

= 1 g

1mEq = 0.0735 g

X mEq = 1 g

X mEq = { (1 x 1) / 0.0735 }

= 13.605 g

44- A 1 miliequivalent (1 mEq) of calcium gluconate = 0.215 g.

How many miliequivalents (mEq) per ml of calcium gluconate a there in a 50 ml vial containing 5 g of calcium gluconate ?

- A. 0.465 mEq/ml
- B. 2.320 mEq/ml
- C. 23.250 mEq/ml
- D. 46.500 mEq/ml

The Answer is (A)

13- A 100 mL infusion bottle contains 2 g of potassium chloride (KCl).
(Mol.wt.kcl= 74.6)
Which of the following is the most likely amount of KCL present in the infusion bottle?

- A. 12.7 mEq
- B. 20 mEq
- C. 26.8 mEq
- D. 78.5 mEq

The answer (C)

$$\text{mEq} = \frac{\text{mg} \times \text{Valence}}{\text{Atomic, formula, or molecular weight}}$$

**A 20 mL vial is labeled potassium chloride (2 mEq/mL). How many grams of potassium chloride (MW= 74.5 g/mol)
are present?**

a) 2.98 g

34- A pharmacist adds 2 g of potassium chloride to 1 L of D5W/1/2 normal saline. Atomic weights (see table).

Sodium	23
Potassium	39
Chloride	35.5
Dextrose	198

Which of the following is the most likely osmolality of solution assuming the final volume as 1 L?

- A. 54 mOsm/L
- B. 300 mosm/L
- C. 405 mOsm/L
- D. 460 mosm/L

Isotonicity?

A patient is prescribed atropine 1% (E=0.13) in purified water and normal saline up to 60ml what is the amount of NaCl required for it to become an isotonic soultion?

a) O. 46 g of Nacl

10- A patient cholesterol level is equal to 4mM/L. This cholesterol level can be expressed in terms of mg/dL

(molecular weight of cholesterol = 386)

A.0.0154 mg/dL

B. 0.154 mg/dL

C. 1.54 mg/dL

D. 15.4 mg/dL

E. 154 mg/dL

$\text{mole} = \dfrac{wt\ ?}{M.wt}$

0,004 386

=1000

wt = 0,004 X 386

$wt = \dfrac{1,544\ g}{L} = \dfrac{\times 1000}{10}$

= 154 gm/dl

$L \xrightarrow{\times 10} dl$

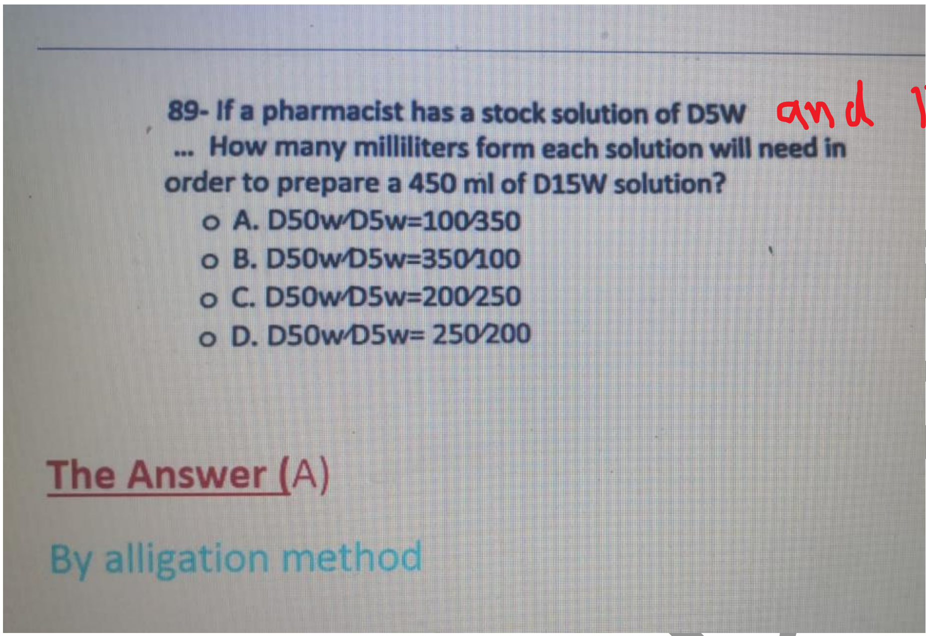
Answer :

Conversion from (mM) to (mg) = conc. × molecular weight

Conversion from (L) to (dL) = conc. / 10

Conc (mg/dl) = conc. (mMol /L) × mwt / 10 = 4×386 /10=154.4

Alligation



and D50

1- sol contain D5W another one contain D50W we want to prepare sol cotain D15W its volune is 450ml … how much ml we need of each sol

a) D50w/D5w=10/35

50		10
	15	
5		35

as ratio D50 : D5

10 : 35

as volume D50 $\frac{10}{45} \times 450$ total vol = 100 ml

$D_5 = 450 - 100 = 350$ ml

or $\frac{35}{45} \times 450 = 350$ ml

2- prescription hydrocortisone 2% Cold cream 60gm You have concentrations of hydrocortisone 2.5% & 1% how many grams will you use from two concentration?

a- 20gm from 1% and 40gm from 2.5%

b- 40gm from 1% and 20gm from 2.5%

c- 30gm from both

answer :

(X) 2.5% ---------- 1 2 – 1 = 1

2%

(Y) 1% ----------- 0.5 2.5 – 2 = 0.5

Complete?

3-How many grams of 5% diclofenac cream should be mixed with 100 g of 1% cream to make a 2.5% diclofenac cream?

A. 30 g

B.60 g

C. 90 g

D. 120

4. How gm of substance X must added to 2000 gm of 10% substance X solution in order to prepare 25% of substance x solution

a) 10000 gm

b) 400 gm

c) 40 gm

d) 10 gm

e) 0.4 gm

answer :

100%		15
	25
10%		75

$\frac{75}{90} \times$ total vol ?? $= 2000$ g

$\frac{15}{90} \times \quad = \boxed{}$

100% -------------- 15 **25 - 10 = 15**

25%

10% ---------------- 75 **100 - 25 = 75**

another way

so the ratio between 100% : 10 % to reach 25% = 15 : 75

2000 gm ------ 75

X gm ------------ 15

X = 2000 x 15 / 75 = 400 gm

5. 30gm of 1% hydrocortisone mixed with 40 gm 2.5% hydrocortisonen what

is the concentration of the resulting solution?

a) 3%

b) 1.85%

c)10%

Answer :

C1.V1 + C2.V2 = C3.V3

30gm × 1% = 0.3gm

40gm × 2.5% = 1gm

So, 1.3 gm is in 70 gm

So, the con. =1.3/70=1.857%

6. if we have 90% of substance X solution , 50% of substance X solution , how mixing both to give 80% of substance X solution ?

a- 3 : 1

b-1:3

c-10:30

d- 5:9

Answer

90%   30

80%

50%   10

So .. 90/50 to reach 80 % equal 30/10 = 3/1

Infusion rate

1. Drug X is a given to a 70 Kg patient at an infusion rate of 0.95 mg/kg/hr.

How much drug we need for a 12-hr infusion bottle

A. **798 mg ….. B.66.5 mg ….. C. 665 mg ….. D. 84 mg**

2-pharmacist calculates a dose of 500 μg/min for continuous infusion of dopamine based on weight of the patient. The concentration of a premixed dopamine infusion is 400 mg/250 ml. What is the most likely amount of dopamine in ml received by the patient in the first hour of treatment?

A. 10 ml
B. 19 ml
C. 24 ml
D. 28 ml

•

52-A solution contains 2.5 mg of a drug per millititre. The drug is to be administered at a rate of 50 mg/hr. (1 mL= 30 drops).
Which of the following is the most appropriate infusion rate?

- A. 5 drops/min
- B. 10 drops/min
- C. 20 drops/min
- D. 40 drops/min

The Answer (B)

Flow rate = (Drop factor X volume)/ Time in minutes

= (30 x 20) / 60 = 10

49-A patient is to receive 750 ml 5% dextrose over five hours. The intravenous infusion set delivers 10 drops\ ml.
Which of the following is the flow rate in ml\min to deliver the prescribed volume ?

- A. 1.5
- B. 2.0
- C. 2.5
- D. 3.0

Clearance: → Cockcroft-Gault equation

1 mmol/L or mg/dL

Kidney note?

Clearance

Male: **97 to 137 mL/min (1.65 to 2.33 mL/s).**

Female: **88 to 128 mL/min (1.496 to 2.18 mL/s).**

5 Stages of Kidney Disease		
	Kidney Function/GFR	Description
Stage 1	> 90%	Normal or High Function
Stage 2	60-89%	Mildly Decreased Function
Stage 3	30-59%	Mild to Moderately Decreased Function
Stage 4	15-29%	Severely Decreased Function
Stage 5	< 15%	Kidney Failure

or on dialysis

Name of cr cl adult : Cockcroft

Name of cr cl children :Schwartz

CrCI< 60 ml/min*
▪ **Drugs may require dosage adjustment**

Inulin?

1. man 40 years and 80 kg sr ce 0.5 mg\dl find creatinie clearance ml/min :

a.222

b.232

Answer :

Cr.cl for male = (140 – age)x weight /72 x ser. Creatinine

=(140 – 40) x 80 / 72 x 0.5 = 222

N.B : The same data for female the answer is : 189

Cr.cl for female = Cr.cl for male x 0.85 = 222 x 0.85 = 188.7

Kg=2.2 pound (ib)

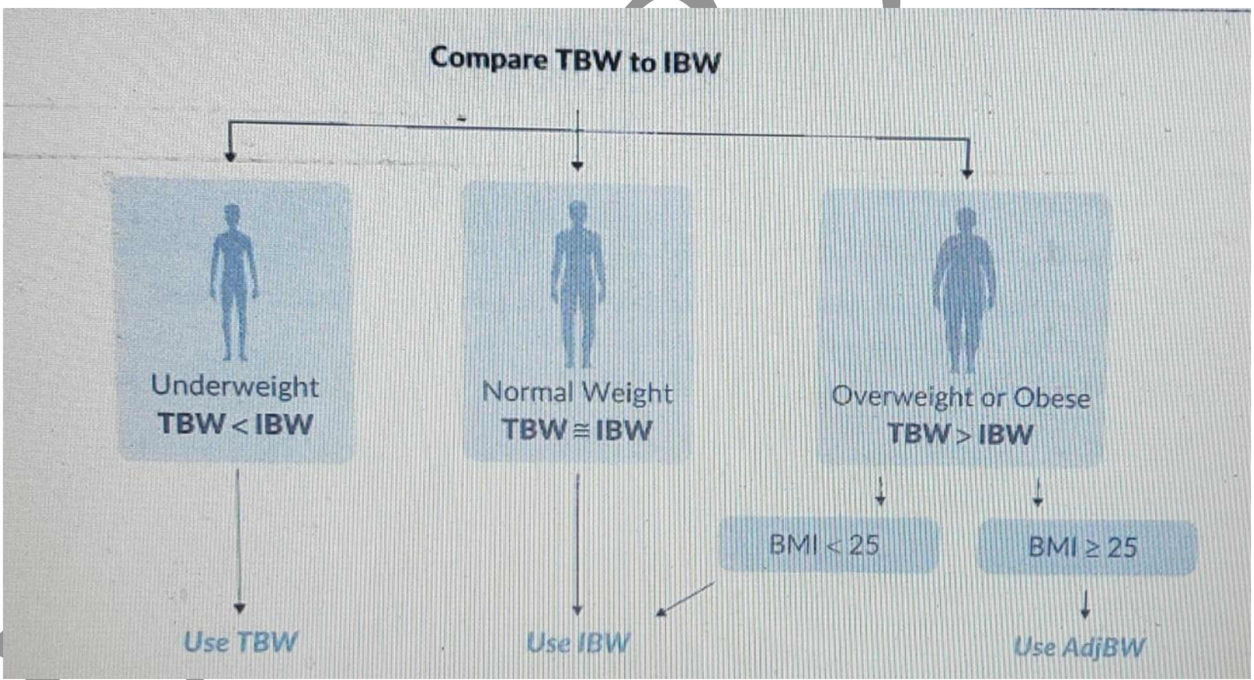
1 inch= 2.54 cm

1 foot = 12 inches

IBW male = 50+2.3 (inch -60) or ht in cm -100 for men

IBW female = 45+2.3(inch -60) or ht in cm – 105 fro female

Adjusted body weight = IBW + 0.4 * (wt - IBW)



- A 64-year-old female patient(height 5'5”, weight 205 pounds) is hospitalizedwith a nosocomial pneumonia which is responding to treatment. Her current antibiotic medications include ciprofloxacin, Primaxin and vancomycin. Her morning laboratory valuesinclude: K 4 mEq/L, BUN 60 mg/dL, SCr 2.7 mg/dL and glucose 222 mg/dL. Based on the renal dosage recommendations from the package labeling below, what is the correct dose of Primaxin for this patient?

Use the flowchart to determine which weight to use to calculate CrCl.

crcl	>_ 71ml/min	41-70ml/min	21-40ml/min	6-20ml/min
Primaxin dose	500mg IVQ6H	500mg IVQ8H	250mg IVQ6H	250mg IVQ12H

Answer:

Use the flowchart to determine which weight to use to calculate CrCl.

Total Body Weight = 93.1818 kg

IBW = 45.5 kg + (2.3 x 5 in) = 57 kg 205 lbs

BMI = $\frac{205\ lbs}{(65\ inches)2}$ x 703 = 34.1 kg/m2, obese

Now calculate her adjusted body weight:

AdjBW04 = 57 + 0.4 (93.1818 - 57) = 71.47 kg

Then, solve using the Cockcroft-Gault equation:

CrCl = $\frac{140-64}{72\ x\ 2.7}$ x 71.47 (0.85) = 23.75 mL/min 72 x 2.7

The correct dose ofPrimaxin is 250 mg IV Q6H.

- Levofloxacin, dosing per pharmacy, is ordered for an 87-year-old female patient (height 5'4", weight 103 pounds). Her labs include BUN 22 mg/dL and SCr 1 mg/dL. Choose the correct dosing regimen based on the renal dosage adjustments from the package labeling below

crcl	>_ 50ml/min	20-49ml/min	<20ml/min
Levofloxacin dose	500mgQ24hours	250mg Q24hours	250mg Q48hours

Answer:

. First, determine which weight to use in calculating the CrCl.

Total BodyWeight = 46.8181 kg

IBW = 45.5 kg + (2.3 x 4 in) = 54.7 kg

Use total body weight for calculating CrCl since the patient's total body weight is less than her IBW.

CrCl = $\frac{140-87}{72\ x\ 1}$ x 46.8181 kg (x 0.85) = 29 mL/min

The correct dose oflevofloxacin is 250 mg Q24H.

0. A 50-year-old male (height 6'1”, weight 177 pounds) has HIV and is being started on tenofovir, emtricitabine and efavirenz therapy. Hislaboratory values include K 4.4 mEq/L, BUN 40 mg/dL, SCr 1.8 mg/dL, and CD4 count of 455 cells/mm3. Using the renal dosage recommendations from the package labeling below, what is the correct dose of tenofovir for this patient?

crc	_> 50 ml/min
Tenofovir dose	300mg daily

Answer:

First, determine which weight to use in calculating the CrCl.

Total Body Weight = 80.4545 kg

IBW = 50 kg + (2.3 x 13 in) = 79.9, or 80 kg

The IBW is almost the same as the actual weight. Either weight will yield a similar CrCl. Next, calculate the CrCl.

140 - 50 CrCl = $\frac{140-50}{72 \, x \, 1.8}$ x 80 kg = 55 mL/min 72 x 1.8

The dose of tenofovir should be 300 mg daily

2- in clinic patient prescriped with a 500mg dose of aspirin , initial plasma conc is 100mg .. With half life 6 hours calculate total body clearance ?

a.0.5 L/hr

b.5 L/hr

c.50 L/hr

Answer:

Vd = dose / initial conc = 500/ 100 = 5L ..... T1-2 = 6 hr

Cl = 0.693 Vd / T1-2 = 0.693 × 5 / 6 = 0.5775 L/hr

3- **aminophylline (80%theophylline) was prescriped for asthmatic patient in a dose of 500mg , half life =6.93 hours how many hours will it take to reach below 2 % ?**

42 hr

Answer:

(80%) ...T1... (40%) ...T2... (20%) ...T3... (10%) ...T4... (5%) ...T5... (2.5%) ...T6... (1.25%)

Time = 6 × T1/2 = 6 × 6.93 = 41.5 hr

4.Drug aminophylline (80% theophylline) in 500ml sln . Half life 6 h .what is the concn of theophylline after 1 day ?

5%

Answer:

1 day = 24 hr = 4 T1-2

(80%) ....T1... (40%) ...T2... (20%) ...T3... (10%) ...T4... (5%)

5. a drug is given as iv infusion in a rate of 2mg/hr ,its T1-2 = 2hr , how much mg of the drug we need to reach steady state

A. 4mg

B. 16mg

C .20mg

D. 40mg

Answer :

We reach steady state after 5 T1-2 = 5 × 2 = 10hr

2mg ...ever... 1 hr

Xmg ...after... 10 hr

X = 2×10/1 = 20mg

6. a drug with T1/2 = 72hr , the body will recive complete dose after ;

A. 1 day

B. 2days

C. 1week

D. 2weeks

Ans: We will reach Steady state after 5 half-life = 5×72= 360hr = 2weeks

7.A problem with the following data

Dose = 1000

Initial conc =10

Elimination rate constant=0.1

Calculate total clearance ??

a-250

b-200

c-150

d-100

e-10 litre

Answer:

Cl= vd × kel

Vd=dose/conc=1000/10=100

Cl=0.1×100=10.

8.A drug with Conc. 400 m and T1/2 = 12 hr.s the concentration will decrease after 1 day by …

a.10%
b.25%
c.75%
d.90%
Answer:

24 hr.s = 2 half lives

(400) ...T1 ... (200) ... T2 ... (100)

so you lose 300 of the drug

(300 / 400) x 100 = 75%

9. 1000 mg of drug follow one compartment.. calculate vd ?

Time	0 hr	2 hrs	4 hrs	6 hrs	12 hrs
Conc	80	58	34	28	10 mg/ml

A.12.5 litre

B. 4 litre

C. 45 litre

Answer :

Vd = dose / initial conc.

Vd = 1000 / 80 = 12.5 L

10.Patient's dose of some drug is 0.5 mg daily and Vd = 500 L .. his body elimination rate is 110.16 Litre per day ... in the last day about 80 % of the drug was in his blood

Calculate half life ..

3 days

Answer:

Cl=0.693 x vd / T1.5

T1/2 = 0.639 x 500 / 110.16 = 3.14 day

Take care about

if Vd per kg

Ionization:

- **If the pH> pKa, more of the acid is ionized, and more of the conjugate base is un-ionized.**
- **If the pH= pKa, the ionized and un-ionized forms are equal.**
- **If the pH< pKa, more of the acid is un-ionized, and more of the conjugate base is ionized**

Which form can cross lipid membrane ?

To calculate the % ionization of a weak acid:

$$\% \text{ ionization} = \frac{100}{1+10^{(pKa-pH)}}$$

To calculate the % ionization of a weak base:

$$\% \text{ ionization} = \frac{100}{1+10^{(pH-pKa)}}$$

A 12-year-old child has bacterial pharyngitis and is to receive an oral antibiotic. Ampicillin is a weak organic acid with a pka of 2.5.what percentage of a given dose will be in the lipid soluble form in the dudenum at pH of 4.5?

A)1%

B)6%

c)9%

d)99%

. What is the percent ionization of amitriptyline, a weak base with a pKa=9.4, at a physiologic pH of 7.4?

A)1%

B)6%

c)9%

d)99%

1..Problem asked to calculate Plasma Osmolarity an you have given some data

Na 140

Glucose=6

Cl 103

Hco3 18

Bun=6

S.cl 8

Answer is : 282.4 osmol

⍰ in general .. to calculate plasma osmolarity follow this equation :

2[Na] +[Glucose]/18 +[BUN]/2.8

correction

$$\text{Corrected phenytoin} = \frac{\text{Phenytoin observed}}{(\text{Albumin} \times 0.2) + 0.1}$$

If cl < 10-15 ml/ min ???? 0.1

Equation for Corrected Calcium

Corrected Ca = Measured Ca + 0.8 * (4 - albumin)

Correction dose

* Vancomycin Correction =

last mintenance dose × $\frac{\text{target level}}{\text{last level}}$

Ex1.is a 72-year-old female (62 kg, 5’6”) with an observed phenytoin level of 6.2 mg/L at 37°C. Her recent albumin level was 2 g/dL, and her serum creatinine level was 0.6 mg/dL. Calculate for the corrected phenytoin level. Crcl = 47.6 ml/min

A.12.4 mg/L

B. 20.6 mg/ L

C. 12.9

patient treated with vancomycin1000mg Q12hr

The current level is 11 and the desired level is 17,

What would the new regimen?

A)1000mgQ12

b)2000mgQ8

c)3000mgQ24

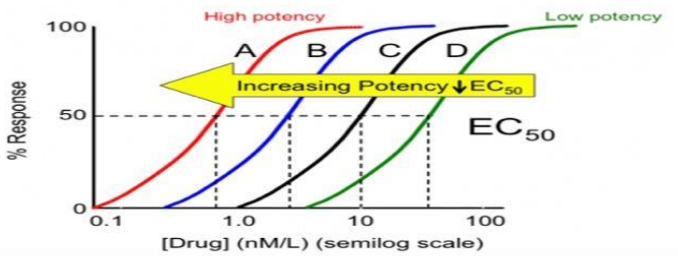
d) 1000mgQ8

$$t_{max} = \frac{\ln(k_a / k)}{k_a - k}$$

Potency and Efficacy

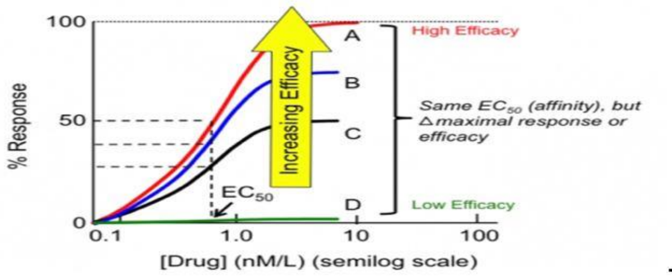
Potency

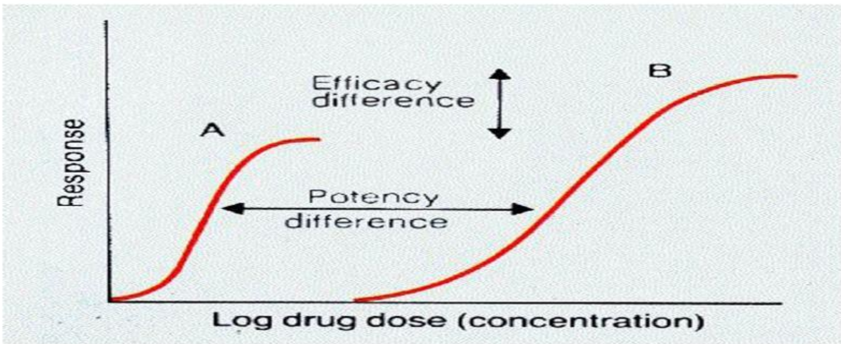
Potency is the concentration (EC_{50}) or dose (ED_{50}) of a drug required to produce 50% of that drug's maximal effect.

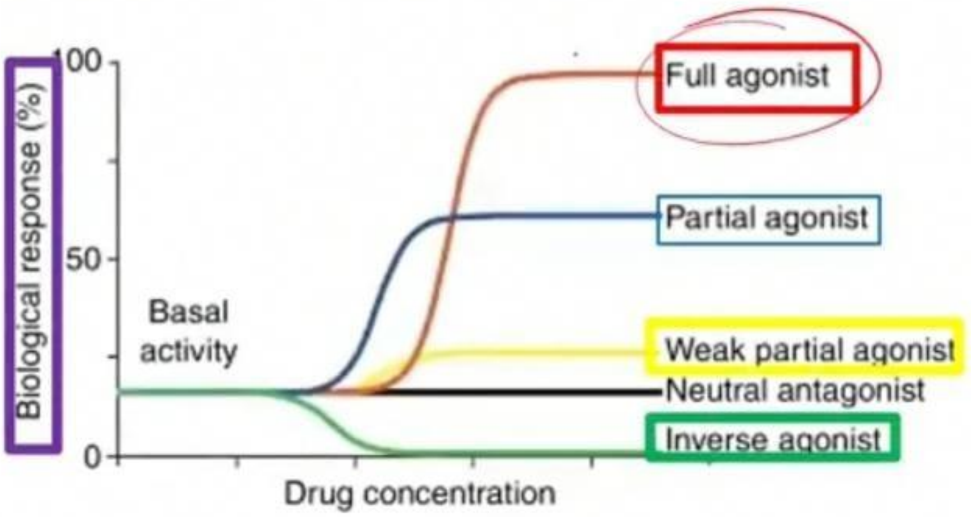


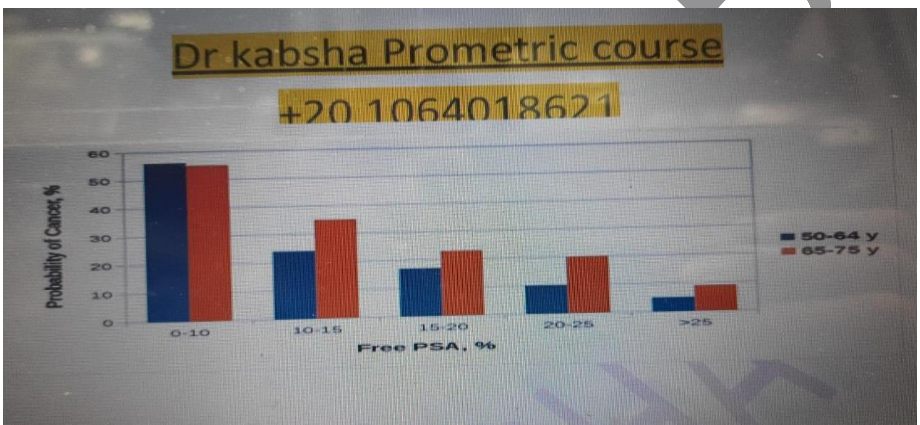
Efficacy

Efficacy (E_{max}) is the maximum effect which can be expected from this drug (i.e. when this magnitude of effect is reached, increasing the dose will not produce a greater magnitude of effect)







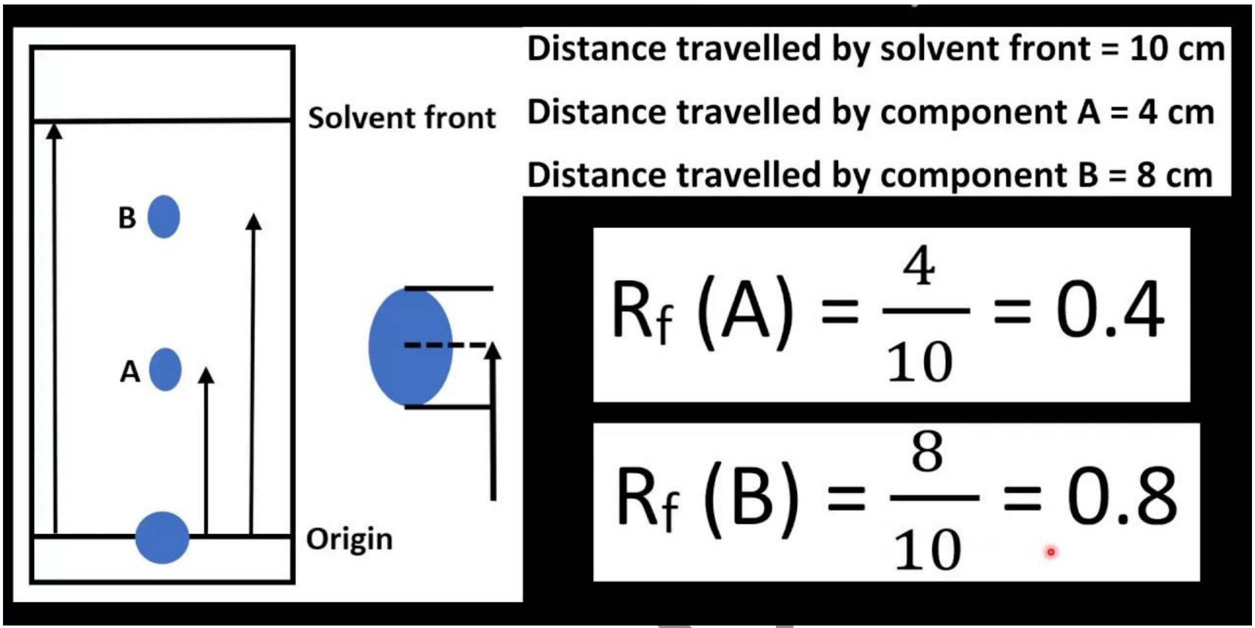


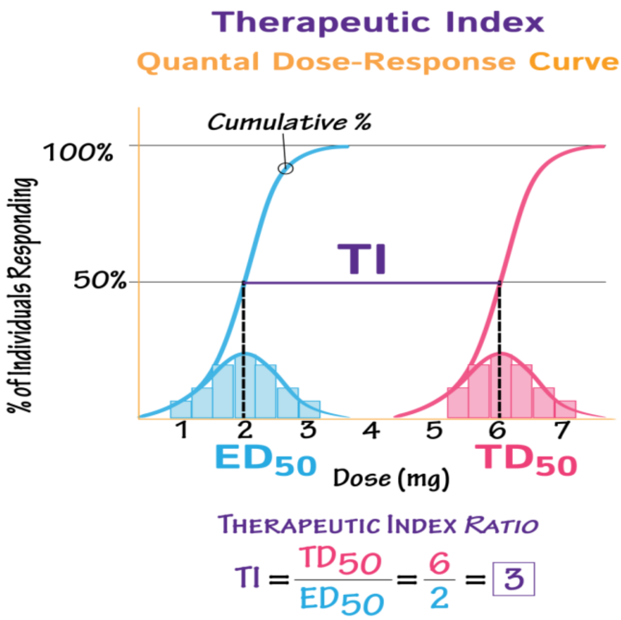
25- Name of this chart?

A- pie chart
B- Histogram
C- forest

DR: KABSHA …PROMETRIC&PEARSON VUECOURSE 00201064018621

Exam **Rf always less than 1**





Anion gap (AG)

Anion gap (AG)= Na- Cl- HCO3

A patient in the ICU has recently developed an acidosis. Using the laboratory parameters below, calculate the patient's anion gap.

Na 139
Cl 101
K 4.6
HCO3 19
SCr 1.6
BUN 38

a)19

85- Voltarol® tablets contain 25 mg of Diclofenac Sodium.

Molecular formula of Diclofenac Sodium: $C_{14}H_{10}Cl_2NO_2$.NA (see table).

	Relative atomic mass
Carbone	12.01
Hydrogen	1.008
Chlorine	35.45
Nitrogen	14.01
Sodium	22.99
Oxygen	16.00

What is the amount of diclofenac per tablet of Voltarol®?

- A. 20.99
- B. 21.98
- C. 23.19
- D. 24.27

The answer (C)

Suppository

Calculate the displacement value of drug If a prescription requires 400 mg of drug per suppository weighing 2 gm. The weight of 6 suppositories with required drug weigh 13.8 g?

Easy calculation by Using formula:

$$DV = \frac{n \times d}{n(b + d) - W}$$

n = Number of suppositories

b = Weight of Base in each suppository

d = Weight of Drug in each suppository

W = Total weight of all suppositories

Weight of Base (b) = 2 g

Weight of Drug in each = 400 mg = 0.4 g

Practical weight (W) = 13.8

$$DV = \frac{6 \times 0.4}{6(2 + 0.4) - 13.8} = \frac{2.4}{0.6} = 4$$

A pharmacist was asked to produce 10 x 1 gram suppositories (each containing 200 mg of drug X). Given that the displacement value Is 4.6, how much base is required?

a)8.4

b)19

Michel minten equation

$$Vo = \frac{V_{max}[S]}{Km + [S]}$$

V0 = Initial velocity (moles/times)
[S] = substrate concentration (molar)
V_{max}= maximum velocity
K_m = substrate concentration at half V_{max}

pt with HTN prescriped nicardipine hydrochloride intravenous infusion at a dose of 5 mg hour for four hours. Nicardipine hydrochloride is available as a 10 ml vial containing 25 mg of nicardipine hydrochloride . How many milliliters of nicardipine should be used ?

A drug vial contains 0.2% of drug X.
Which of the following is the amount of the drug X (in mg) in a 5 ml dose?
5 ml dose?

A. 0.1
B. 1
C. 10
D. 100

A solution contains 27 mg of antipyrine per 500 microliter.
How many grams of antipyrine are there in 3 ml of the solution?

A. 0.162 gm
B. 0.270 gm
C. 162 gm
D. 270 gm

A solution contains 5 gm of drug per 25 ml.
Which of the following is the concentration, in mg/ml, of the drug?

o A. 50 mg/ml
o B. 100 mg/ml
o C. 200 mg/ml
o D. 400 mg/ml

How many grams of dextrose are in a 250 ml bag of 10% dextrose?

o A. 12.5
o B. 25
o C. 50
o D. 100

A 20 ml vial of penicillin-G potassium containing 2,000,000 units is available. Each mg of the drug is equivalent to 1600 units.
How many milligrams of penicillin-G potassium are there in one ml of the solution?

o A. 12.50
o B. 62.50
o C. 160
o D. 1250

Henry's law is a gas law which states that at the amount of gas that is dissolved in a liquid is directly proportional to the partial pressure of that gas above the liquid when the temperature iskept constant. The constant of proportionality for this relationship is called Henry's law constant(usually denoted by 'k_H'). The mathematical formula of Henry's law is given by:

$P \propto C$ (or) $P = k_H.C$

Where,

- **'P' denotes the partial pressure of the gas in the atmosphere above the liquid.**
- **'C' denotes the concentration of the dissolved gas.**
- **'k_H' is the Henry's law constant of the gas.**

The

Acid Base (the Metabolic component IV)

The Van Slyke equation
(= The equation for the CO2 equilibration curve of blood *in vitro*)

$$a-24.4 = -(2.3*b+7.7)*(c-7.40)+d/(1-0.023*b)$$

where:
a = bicarbonate concentration in plasma (mmol/l)
b = hemoglobin concentration in blood (mmol/l)
c = pH of plasma at 37 oC
d = base excess concentration in blood (mmol/l)

(Scand J Clin lab Invest 1977:37,Suppl 146:15-20)

The Henderson-hasselblach equation can be used to calculate the amount of acid and conjugatebase to be combined for the preparation of a buffer solution having a particular PH

Noyes whitney ……dissolution rate

Fick's law ……… rate of diffusion

Michel minten equation…….. velocity

Facebook group

https://www.facebook.com/groups/477348776664047/?ref=share

YouTube channel

https://youtube.com/@kabshaacademy4741

Telegram group

https://t.me/joinchat/TBSn0BhPMtU8TOasI8Q7Mg